U-Net with Attention
and
Evaluation Metric for Segmentation

What is U-Net with Attention

- ✓ Attention U-Net is an attention gate (AG) model for medical imaging that automatically learns to focus on target structures of varying shapes and sizes.

- ✓ Models trained with AGs implicitly learn to suppress irrelevant regions in an input image while highlighting salient features useful for a specific task.

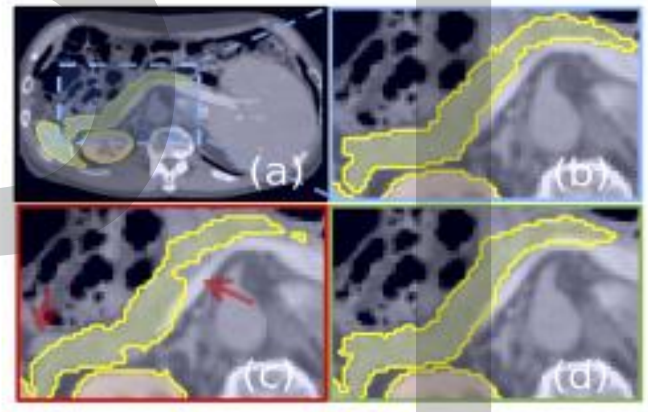
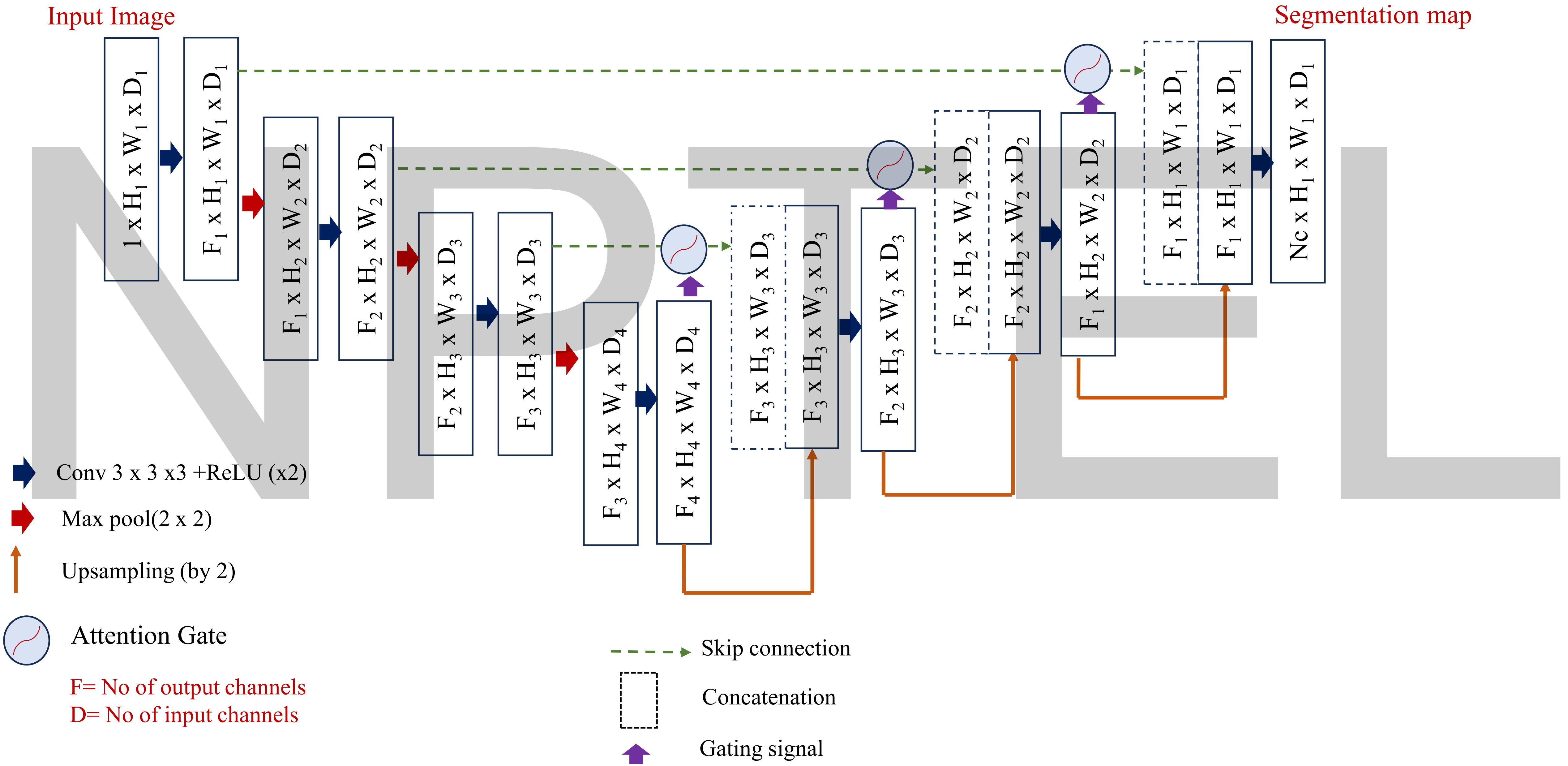


Figure 3(b): The ground-truth pancreas segmentation (a) is highlighted in blue (b). Similarly, U-Net model prediction (c) and the predictions obtained with Attention U-Net (d) are shown. The missed dense predictions by U-Net are highlighted with red arrows.

Ref:

Attention U-Net: Learning Where to Look for the Pancreas; Ozan Oktay.,et al; arXiv:1804.03999 (11th April, 2018)

Attention U-Net Segmentation Model



Attention Gate (AG) Mechanism

Core Idea

Instead of directly concatenating encoder features to the decoder features, we first **weight** the encoder features with an **attention coefficient** (a number between 0 and 1) for each spatial location.

g → W_g: 1 x 1 x 1

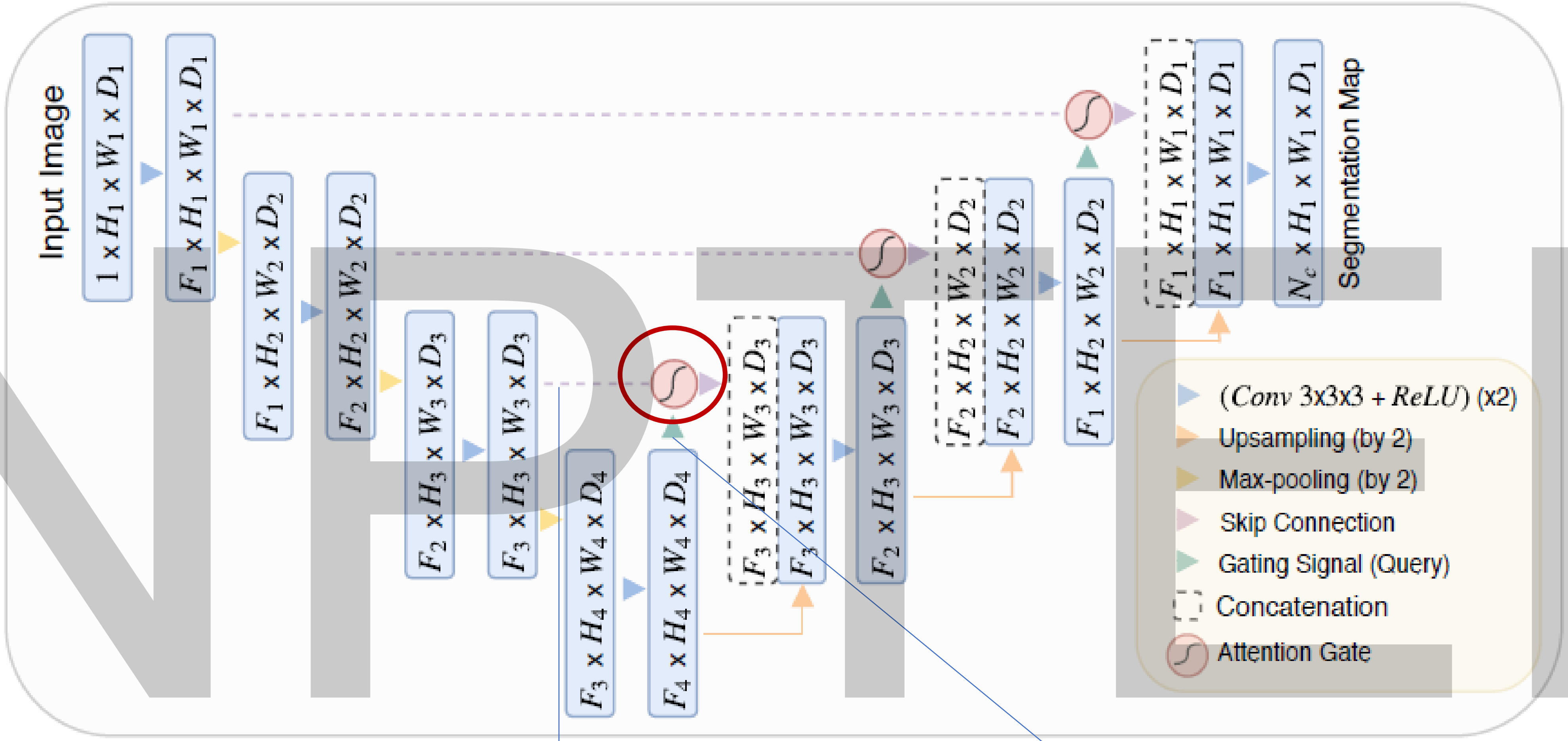
Gating signal

- ✓ Gating signal "g" comes from decoder.
- ✓ Contains textual information but in coarser scale
- ✓ Low-resolution, deeper feature map from decoder

x^l → W_x: 1 x 1 x 1

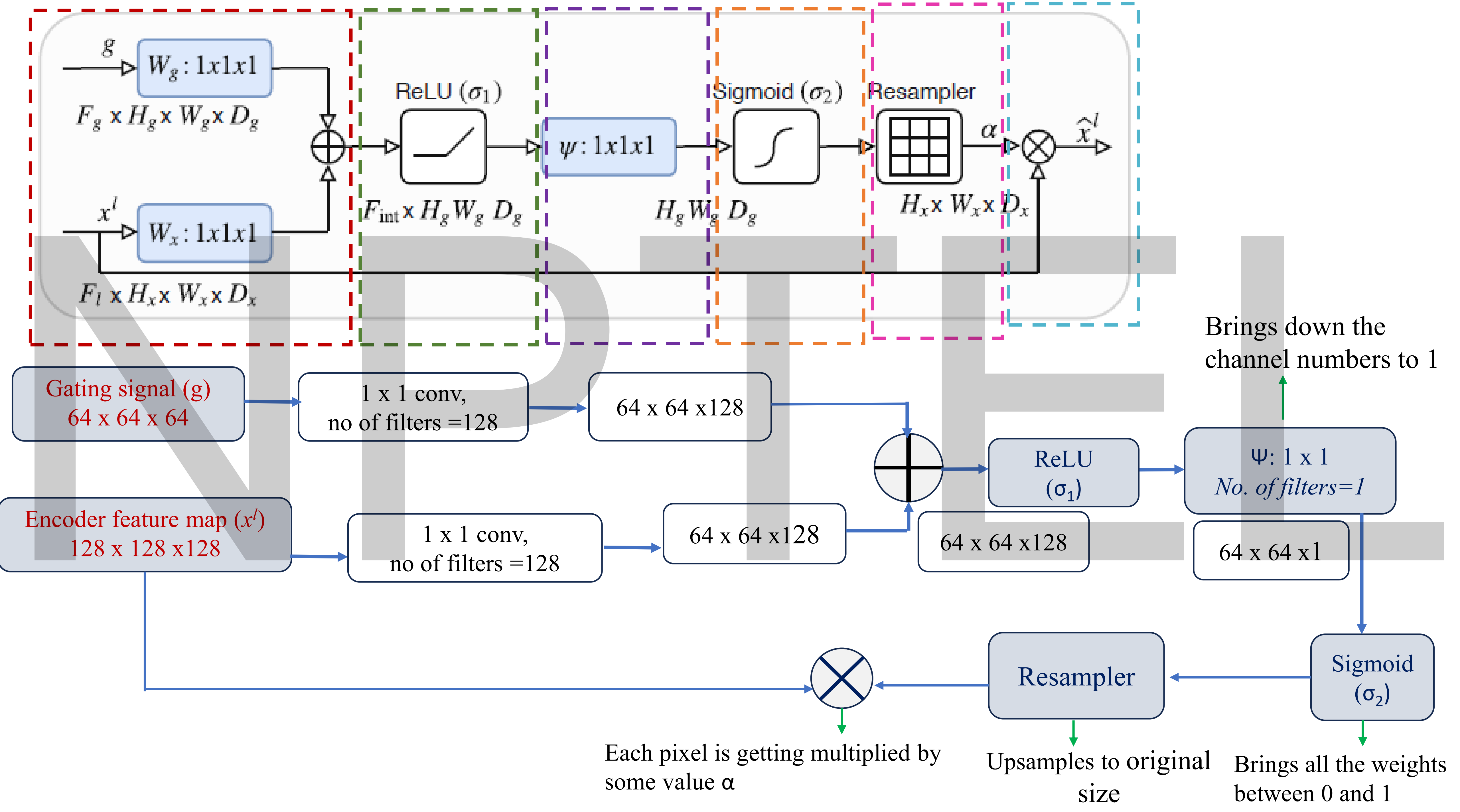
Encoder feature map of layer l

- ✓ Feature map from the encoder (skip connection).
- ✓ High resolution



Encoder feature map
128 x 128 x128

Gating signal (g)
64 x 64 x 64



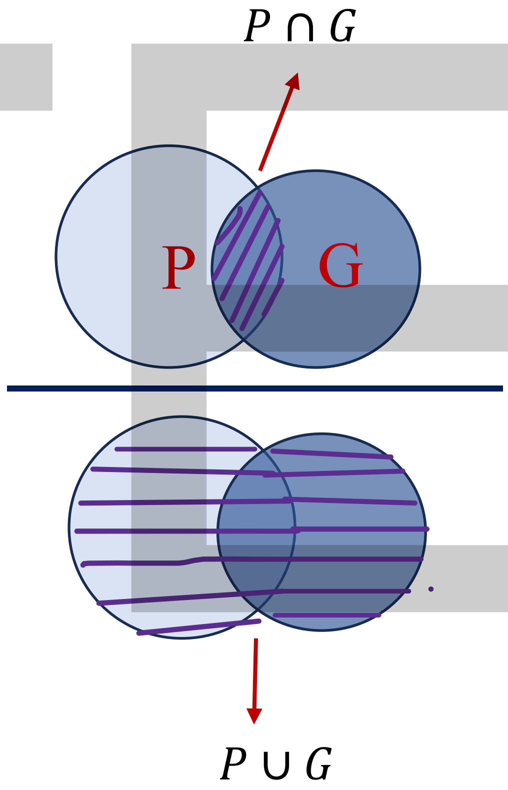
g
W_g : 1x1x1
F_g x H_g x W_g x D_g
x^l
W_x : 1x1x1
F_l x H_x x W_x x D_x
ReLU (σ_1)
F_{int} x H_g W_g D_g
ψ : 1x1x1
H_g W_g D_g
Sigmoid (σ_2)
Resampler
α
H_x x W_x x D_x
$\hat{x}^l$
Brings down the channel numbers to 1
Gating signal (g)
64 x 64 x 64
1 x 1 conv,
no of filters =128
64 x 64 x128
Encoder feature map (x^l)
128 x 128 x128
1 x 1 conv,
no of filters =128
64 x 64 x128
ReLU
(σ_1)
64 x 64 x128
Ψ: 1 x 1
No. of filters=1
64 x 64 x1
Sigmoid
(σ_2)
Resampler
Each pixel is getting multiplied by some value α
Upsamples to original size
Brings all the weights between 0 and 1

Evaluation metrics for Image Segmentation

Intersection over Union (IoU)

Also known as Jaccard Index

$$\text{IoU} = \frac{|\, P \cap G \,|}{|\, P \cup G \,|}$$

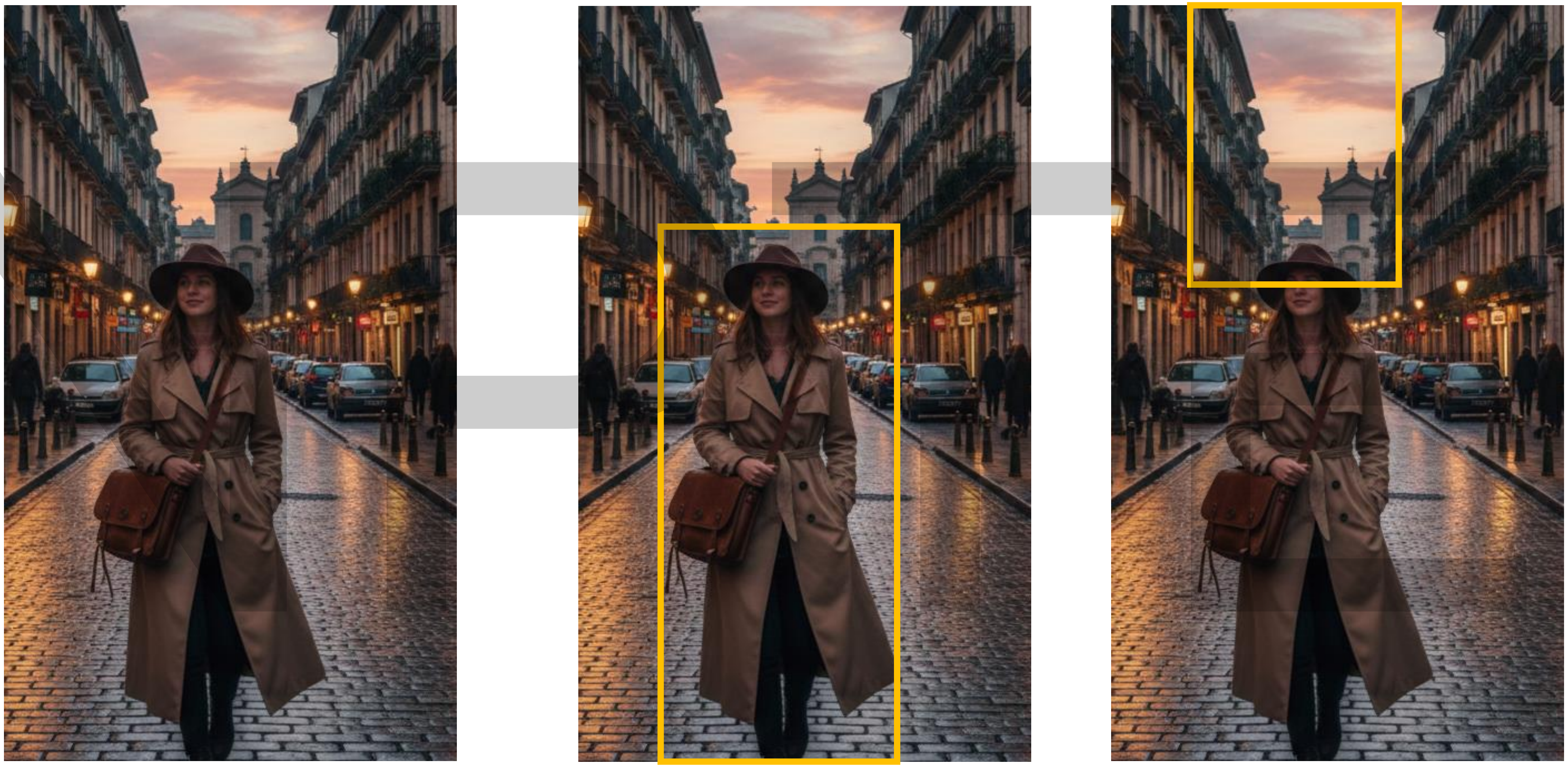


Where:

- ✓ P= predicted mask
- ✓ G= ground truth mask
- ✓ $P \cap G$= pixels correctly predicted as the object (True Positives)
- ✓ $P \cup G$= all pixels predicted *or* actually belonging to the object

Range: 0 to 1 (higher = better)

True Positive and False Positive in object detection



True Positive (TP)

False Positive (FP)

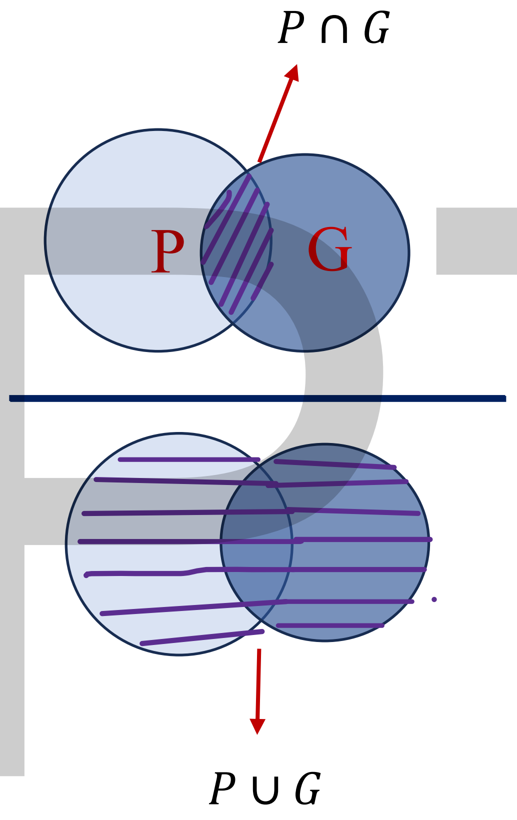
True Negative and False Negative in object detection

True Negative (TN)

False Negative (FN)

IOU in terms of TP, TN, FP and FN

IoU =

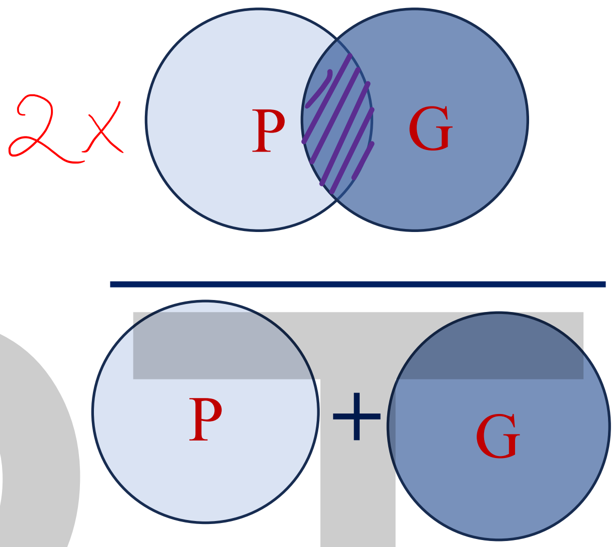


$$= \frac{TP}{TP+FP+FN}$$

For semantic segmentation prediction, the IoU score is calculated for each class separately and then **averaged over all classes** to provide a global, mean IoU score.

Dice Similarity Coefficient (DSC)

$$\text{Dice} = \frac{2|P \cap G|}{|P|+|G|}$$



$$= \frac{2TP}{2TP+FP+FN}$$

Dice similarity coefficient is often used in medical image segmentation for very small or thin structures.

Pixel Accuracy (PA)

$$Pixel\ accuracy = \frac{\text{correctly predicted pixels.}}{\text{total number of pixels}}$$

$$= \frac{TP+TN}{TP+TN+FP+FN}$$

Precision and Recall

Precision tells you out of all the pixels which the model marked as positive, how many were actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall tells you out of all the actual positive pixels, how many the model successfully found.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Example:
If the model says 100 pixels are class-1:
80 are correct → TP
20 are wrong → FP
Precision = 80 / (80 + 20) = 0.80

If the model missed 40 actual class-1 pixels → FN
Then:
Recall = 80 / (80 + 40) = 0.67

NPTEL

Thank you